

SalbaBayan



A flood and storm warning app for one barangay — built to keep working when the mobile network does not.

The warning arrives exactly when the network leaves.

A barangay's storm warning travels by megaphone, text message and group chat. All three fail in the same hour: the towers saturate, the power goes, and the people who most need to move are the ones who hear nothing.

Residents

Do not know the current signal, which centre is nearest, or whether the road there is passable.

Volunteers

Take rescue calls by phone and radio, with no shared list and no way to see who is already going.

Officials

Have no live picture of their own barangay — which streets are flooded, how deep, or how many are in the hall.

Assume the network is gone. Design from there.

Every other design decision in SalbaBayan follows from one: nothing a person does in a storm may depend on a working connection, and nothing on the screen may claim to be current when it is not.

Written down first

An SOS, a report, a headcount is saved on the phone in milliseconds and sent when the network returns. The tap never waits on a tower.

The map is on the phone

The barangay's streets, outline, routes and wording are cached. With no signal the app draws its own map and routes on it.

Honest about age

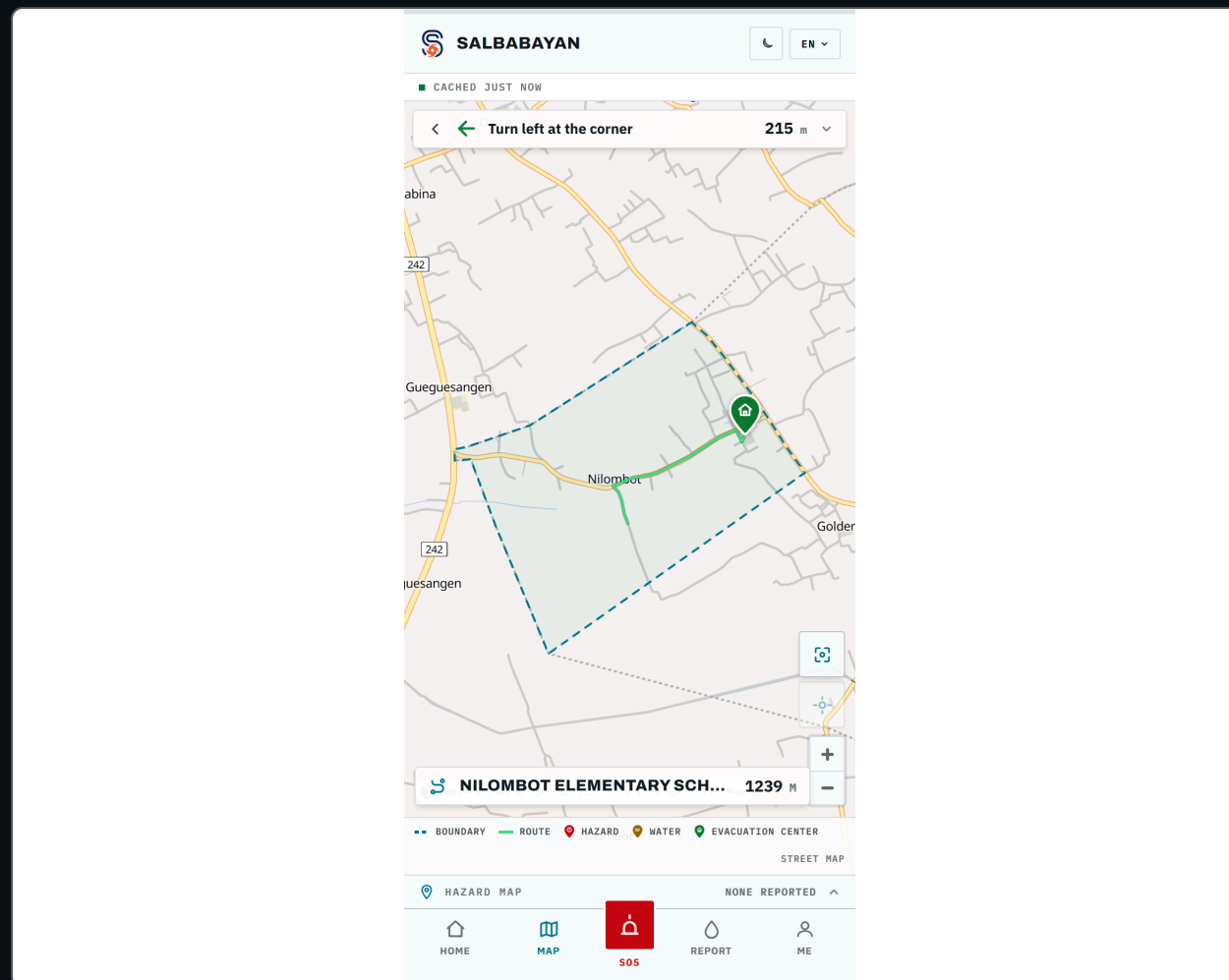
"Cached 2 hours ago" is on screen wherever it is true. A cache that forges freshness is worse than no cache at all.

WHAT A RESIDENT GETS

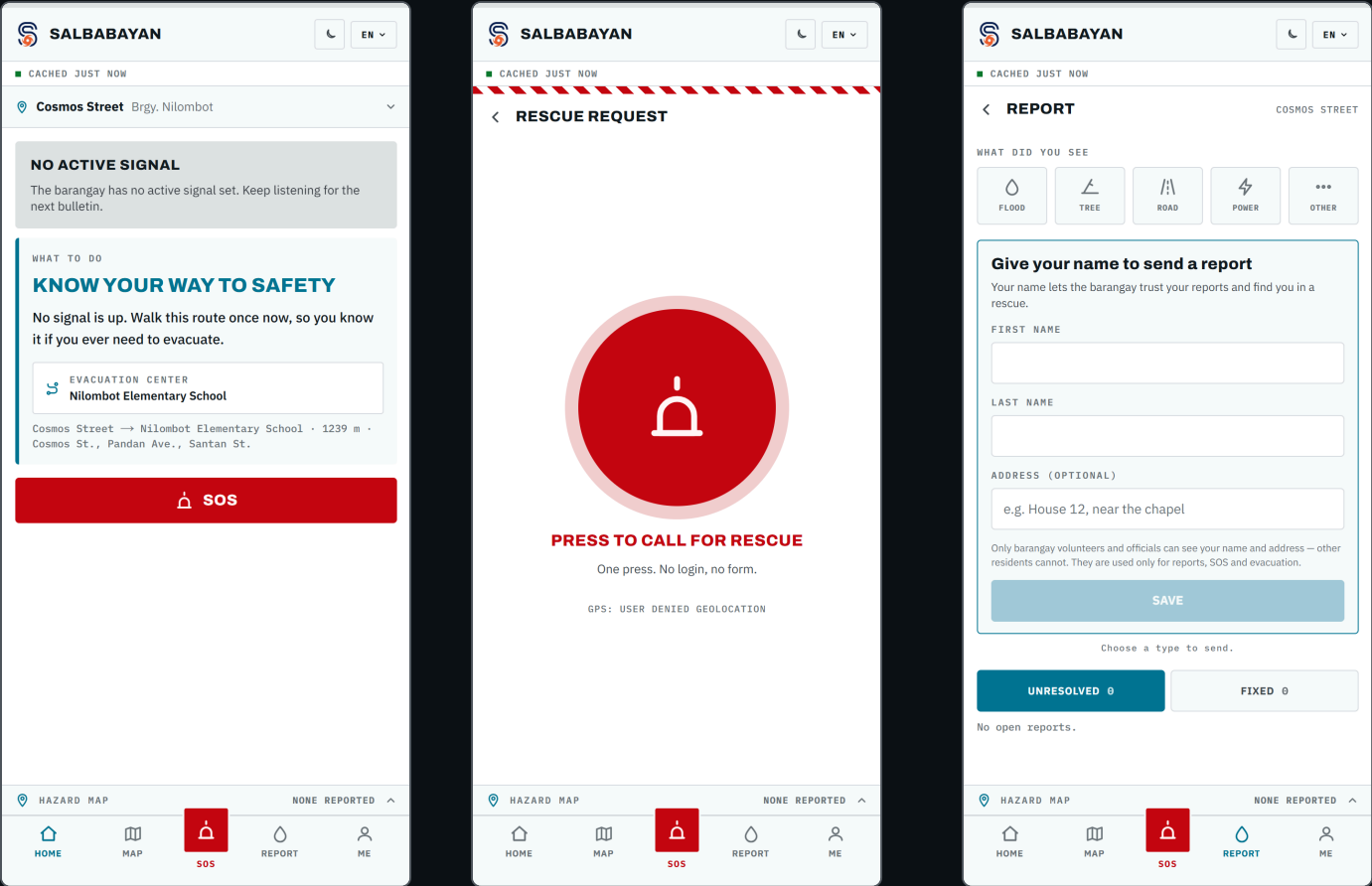
The signal, the route, and a way to be found.

- **The barangay's signal** and what it means for their own street, with the leave-before time counting down.
- **A walking route** to the nearest of three evacuation centres, computed on the phone from the streets — not a straight line.
- **Hazards and flooding** reported by anyone, on one map, with old readings faded rather than hidden.
- **One-press SOS.** No login, no form, no waiting for GPS.

No account, no email address, no telephone number. The app opens as a resident of this barangay the moment it loads.



Four screens, and the one that matters is a button.



HOME – THE SIGNAL AND WHAT TO DO SOS – ONE PRESS, WORKS OFFLINE REPORT – FLOOD DEPTH BY BODY HEIGHT

Flood depth is reported against the body – knee, waist, chest, above head – because nobody in a storm is estimating centimetres.

A shared queue, and a private route to the call.

1 • Raised

The resident presses once. The call is timed from the press, even with no signal, and sent the moment there is one.

2 • Seen

Any volunteer acknowledges it. The resident's screen changes to SEEN BY THE BARANGAY with the waiting time.

3 • Taken

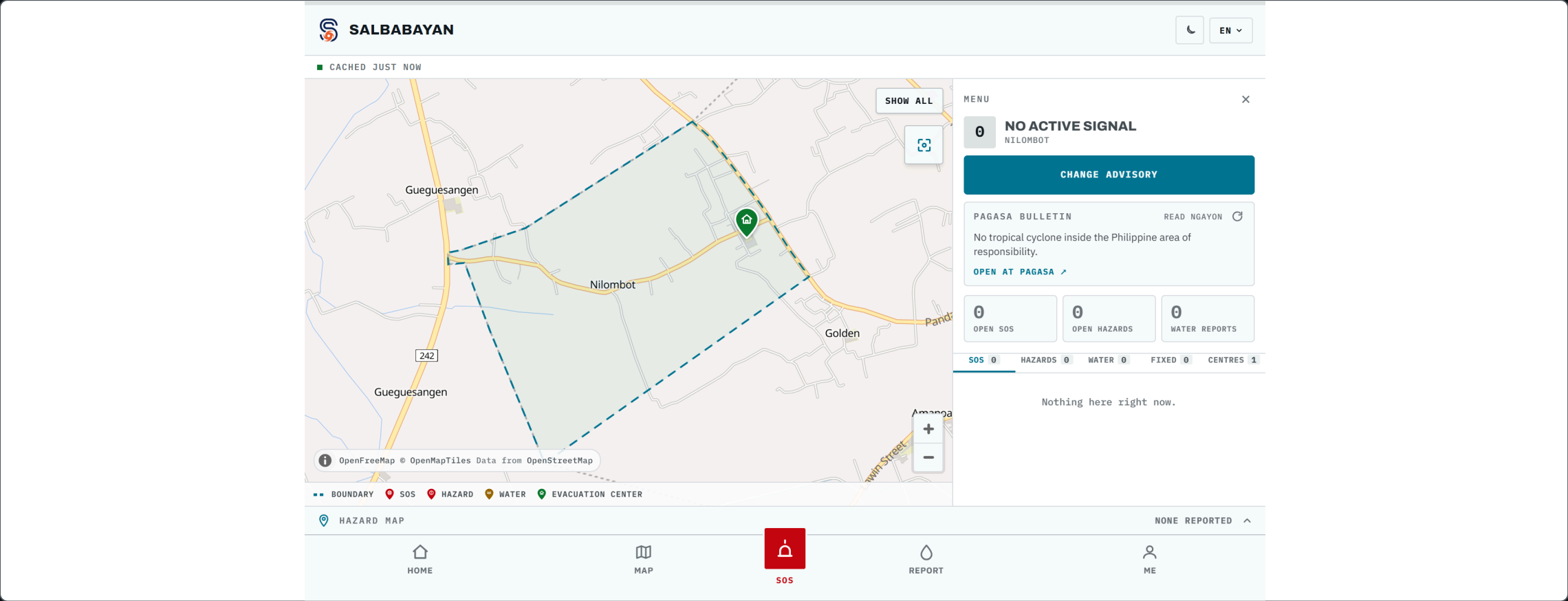
Each responder gets their own walking route from where they are. Several may take the same call; none sees another's line.

4 • Rescued

Marking it closes the call for everyone and clears the route from every phone that was going.

The rescue map carries the hazards and the flood readings too, so a volunteer can see the blocked road before setting out — not on arriving at it.

The whole barangay, on one screen.



Tap a street to place a hazard, a flood depth or an evacuation centre · the advisory and the PAGASA bulletin sit together in the sidebar · every open SOS, hazard and reading is on the map.

The national bulletin, read for this barangay.

PAGASA publishes no API, so the dashboard reads the tropical cyclone bulletin itself and works out **which wind signal covers this barangay** — then fills the advisory form on request.

- **It fills the form. It never issues.** An official reads every field and holds the confirm control, as when they type it.
- **It refuses to guess.** "The western portion of Pangasinan" names the province without covering Mapandan, so the signal is shown and the form left alone.
- **It fails loudly.** If the bulletin cannot be read, the card says so and links to PAGASA.

Why a human stays in the loop

PAGASA's signal covers a province. The barangay's signal is a decision about this barangay, made by someone who knows which streets flood first and how long it takes to walk out of them.

A misread bulletin is then a wrong suggestion on one screen — never a wrong signal on every phone in the barangay.

Who can do what, enforced by the database.

RULE	HOW IT IS ACTUALLY ENFORCED
Roles are granted	A phone is a resident until an official grants it more, by device code. Nobody can promote themselves — the function that once allowed it is gone.
Official pages	A 4-digit PIN each session, on top of the role. A phone that is not an official in the database cannot open them whatever it types.
Names	Visible to volunteers and officials only, never to other residents. The SOS button never asks for one.
Every read and write	Row-level security in Postgres, checked by a suite that runs against the live database on every build — not by the app being polite.
Flood readings	Only officials may clear or overwrite one, so the map cannot be quietly emptied by a passer-by.

The MVP is live, and it is checked.

432

AUTOMATED CHECKS, EVERY BUILD

41

OF THOSE AGAINST LIVE DATABASE
SECURITY

40

LANGUAGES OFFERED • 3 WRITTEN
BY PEOPLE

4,411

RESIDENTS IN THE BARANGAY IT IS
BUILT FOR

WORKING NOW

- Advisory, protocols and the countdown, offline
- Routing to the nearest of three centres, computed on the phone
- SOS end to end, multi-responder, with per-responder routes
- Hazard and flood reporting, two-way between residents and officials
- QR check-in and an append-only headcount ledger
- The PAGASA bulletin on the official dashboard
- Push alerts that reach a volunteer with the app closed

NEXT

- **Pangasinan and Ilocano written by native speakers** — Mapandan speaks Pangasinan, Ilocano is widely spoken across the province, and the app machine-translates both
- **Waterways on the offline map** — a flood app whose offline map has no rivers
- **The parser against a real bulletin** — no cyclone has entered the area since it was built
- **A rehearsal with the barangay** before the season

DELIVERABLES

Links

FINAL MVP

<https://salba-bayan.vercel.app>

SOURCE CODE

<https://github.com/JOHNM-GOAT/SalbaBayan>

DELIVERABLES

github.com/JOHNM-GOAT/SalbaBayan/tree/main/stage-5

IN THAT FOLDER

SalbaBayan-Pitch-Deck.pdf · SalbaBayan-User-Manual.pdf

Open the MVP on a phone, turn airplane mode on, and reload it.

That is the whole argument.